A nonlinear Newton time cell: controlled fluctuations and induced stiffness

Result, 2026-10-27; written derivation, unrefereed. An arbitrary intermediate time can be eliminated exactly from a positive Euclidean mechanical weight. For a convex potential, the normalized nonlinear conditional law has an explicit fluctuation bound, and the retained interaction has bounded curvature even when the original force derivative is unbounded. Both integration and classical minimization soften the curvature. They differ at a quartic flat centre: positive supplied $\hbar$ generates strictly positive quadratic curvature, while classical elimination retains zero quadratic curvature.

This is a concrete $0+1$ use of Balaban's instruction to retain the full normalized interaction and its large-field contribution. It supplies a nonlinear calculation beyond the Gaussian reference of the time-blocking note. The positive weight and its action multiplier are premises. An experimentally inferred positive curvature would need additional arguments before being identified with a necessary universal quantum action.

Targeted check (GPT-6.1 Terra, 2026-10-27). Terra read this note and checked the normalized moments/tails, variance and curvature identities, quartic comparison and curvature constants, and one-halving conditional independence without finding a mathematical error under the stated hypotheses. This is a targeted written check; no independent validation of the physical weighting is claimed.

The Gaussian-convolution and quadratic-infimum constructions are standard; we prove the cell statements below and claim no literature novelty. Source motivation: Dimock I, sections 2.1, 4.5 and Theorem 14, and Dimock II, Theorem 3.1 and sections 3.13–3.14 (selected passages read). The old quartic limit theorem, section 3, uses $V=\lambda q^4$; here the convention is $V=\lambda q^4/4$. The two normalizations must not be mixed.

1. The complete one-vertex weight

Let $m,u,v,\hbar>0$, $T=u+v$, and fix positions x,y. Put

$$\mu=\frac{vx+uy}{T},\qquad a=\frac{mT}{uv},\qquad w=\frac{T}{2}. \tag{1}$$

Here a has units of action per length squared and w is a time. Completing the kinetic square gives

$$\frac{m(z-x)^2}{2u}+\frac{m(y-z)^2}{2v}=\frac{m(y-x)^2}{2T}+\frac{a}{2}(z-\mu)^2. \tag{2}$$

The symmetric Euclidean action also includes $uV(x)/2+wV(z)+vV(y)/2$. Define

$$I_\hbar(\mu)=\sqrt{\frac{a}{2\pi\hbar}}\int_{\mathbb{R}}\exp\left[-\frac{F_\mu(z)}{\hbar}\right]dz,\quad F_\mu(z)=\frac{a}{2}(z-\mu)^2+wV(z),\quad A_\hbar=-\hbar\log I_\hbar. \tag{3}$$

Then the integrated two-cell kernel, with the free normalization at each cell, is exactly

$$\sqrt{\frac{m}{2\pi\hbar T}}\exp\left[-\frac{1}{\hbar}\left\{\frac{m(y-x)^2}{2T}+\frac{u}{2}V(x)+\frac{v}{2}V(y)+A_\hbar(\mu)\right\}\right]. \tag{4}$$

We retain the whole function $A_\hbar$; it is not generally a quartic polynomial. The normalized conditional law $\nu_{\hbar,\mu}$ of Z is $e^{-F_\mu/\hbar}dz$ divided by its integral. No unnormalized action cost is used as a probability estimate.

For comparison define

$$A_0(\mu)=\min_z F_\mu(z). \tag{5}$$

This is a classical Euclidean stationary elimination. It is not the real-time Newton action: for real time the potential sign in (2)–(4) changes and an oscillatory integral replaces the positive weight. In particular, these probability bounds cannot simply be continued to real time. No c or relativistic premise is used here.

2. Convexity gives normalized control

Assume $V \in C^2(\mathbb{R})$ is convex, with sufficient growth regularity for the following integrations by parts. Polynomials with nonnegative second derivative satisfy the hypotheses. The Gaussian kinetic term makes F_μ strongly convex, so it has one minimizer $z_*(\mu)$ and $F''_\mu \geq a$. All moments needed below are finite.

Proposition 1. For $n = 1, 2, \ldots$ and $r > 0$,

$$\mathbb{E}_{\nu_{\hbar,\mu}}(Z - z_*)^{2n} \leq (2n-1)!! \left(\frac{\hbar}{a}\right)^n, \qquad \nu_{\hbar,\mu}(|Z - z_*| \geq r) \leq 2\exp\left[-\frac{ar^2}{2\hbar}\right]. \tag{6}$$

Also

$$\mathrm{Var}(Z) \leq \frac{\hbar}{a}, \qquad A'_\hbar = a(\mu - \mathbb{E}Z), \qquad A''_\hbar = a - \frac{a^2}{\hbar}\,\mathrm{Var}(Z), \qquad 0 \leq A''_\hbar \leq a. \tag{7}$$

Proof. Write $Y = Z - z_*$. Strong convexity and $F'_\mu(z_*) = 0$ imply $YF'_\mu(z_* + Y) \geq aY^2$. Integration by parts gives

$$\mathbb{E}[Y^{2n-1}F'_\mu(Z)] = (2n-1)\hbar\mathbb{E}Y^{2n-2}. \tag{8}$$

The left side is at least $a\mathbb{E}Y^{2n}$. Induction proves the moment bound. Expanding $\cosh(tY)$ in its nonnegative even powers gives $\mathbb{E}\cosh(tY) \leq e^{\hbar t^2/(2a)}$. Since $e^{t|Y|} \leq e^{tY} + e^{-tY}$, exponential Markov inequality with $t = ar/\hbar$ gives the tail bound.

For variance, take independent copies Z, Z'. Integration by parts also gives $\mathbb{E}F'_\mu = 0$ and $\mathrm{Cov}(Z, F'_\mu(Z)) = \hbar$. Consequently

$$\hbar = \frac{1}{2}\mathbb{E}[(Z - Z')(F'_\mu(Z) - F'_\mu(Z'))] \geq a\,\mathrm{Var}(Z). \tag{9}$$

Differentiating the normalized integral (3) gives the two derivative identities in (7); in particular $\partial_\mu \mathbb{E}Z = a\,\mathrm{Var}(Z)/\hbar$. This proves the curvature bounds. □

If additionally $0 \leq V'' \leq K$, the same two-copy identity with the upper slope bound gives

$$\frac{\hbar}{a + wK} \leq \mathrm{Var}(Z) \leq \frac{\hbar}{a}, \qquad 0 \leq A''_\hbar \leq \frac{awK}{a + wK}. \tag{10}$$

These are bounds for the full normalized nonlinear conditional law. They require neither a quadratic approximation nor a small quartic coupling.

The function $A_\hbar/w$, viewed as a potential of the coordinate μ, has force derivative bounded by $a/(mw) = 2/(uv)$. This is a mathematically well-defined coarse potential. Identifying its gradient with the force of a physical Newtonian body is a further modeling choice: (4) is an endpoint kernel, not an autonomous Newton equation derived from it.

3. Classical elimination has the same curvature bound

The minimizer satisfies

$$a(z_* - \mu) + wV'(z_*) = 0, \qquad z'_* = \frac{a}{a + wV''(z_*)}. \tag{11}$$

The implicit function theorem and direct differentiation yield

$$A_0' = a(\mu - z_*), \qquad A_0'' = \frac{awV''(z_*)}{a + wV''(z_*)}, \qquad 0 \le A_0'' \le a. \tag{12}$$

Thus bounded coarse curvature is already possible in classical elimination. For the convex quartic $V = \lambda z^4/4$, $\lambda > 0$, the positive-$\hbar$ interaction approaches this classical envelope with an explicit bound:

$$0 \le A_\hbar - A_0 \le \frac{3w\lambda z_*^2\hbar}{2a} + \frac{3w\lambda\hbar^2}{4a^2}. \tag{13}$$

Proof. After centering the integral at z_*, let $Y \sim N(0, \hbar/a)$ and $R_V = V(z_* + Y) - V(z_*) - V'(z_*)Y \ge 0$. Cancellation of the linear terms in F_μ gives

$$A_\hbar = A_0 - \hbar \log \mathbb{E} e^{-wR_V/\hbar}. \tag{14}$$

The expectation is at most one, and Jensen's inequality bounds the additional action by $w\mathbb{E}R_V$. For the quartic, the Gaussian second and fourth moments give exactly the right side of (13). This proves local uniform convergence as $\hbar \downarrow 0$. □

Classical minimization and positive integration therefore share one regularity mechanism. A criterion requiring only a finite ODE Lipschitz constant would not distinguish them in this convex cell.

4. The quartic flat centre does distinguish the two weights

For $V = \lambda z^4/4$ and $\mu = 0$, symmetry gives $\mathbb{E}Z = 0$ and $z_* = 0$. Write

$$g = \frac{w\lambda\hbar}{a^2}, \qquad \chi_\hbar = A_\hbar''(0). \tag{15}$$

The classical curvature is $A_0''(0) = 0$. For every supplied $\hbar > 0$, however,

$$0 < a\frac{\sqrt{1+4g} - 1}{\sqrt{1+4g} + 1} \le \chi_\hbar \le a\min(1, 3g). \tag{16}$$

Moreover $\chi_\hbar/(ag) \to 3$ as $g \downarrow 0$.

Proof. Integration of $(ze^{-F_0/\hbar})'$ gives

$$a\mathbb{E}Z^2 + w\lambda\mathbb{E}Z^4 = \hbar, \qquad \chi_\hbar = \frac{aw\lambda}{\hbar}\mathbb{E}Z^4. \tag{17}$$

Let $r = a\mathbb{E}Z^2/\hbar$. Jensen's inequality $\mathbb{E}Z^4 \ge (\mathbb{E}Z^2)^2$ gives $r + gr^2 \le 1$, hence $r \le 2/(1 + \sqrt{1+4g})$. Equation (7) gives $\chi_\hbar = a(1-r)$, proving the lower bound. Equation (6) at $n = 2$ gives $\mathbb{E}Z^4 \le 3(\hbar/a)^2$, proving the upper bound together with $\chi_\hbar \le a$.

Under $Z = \sqrt{\hbar/a}\,X$, the normalized density of X is proportional to $e^{-X^2/2 - gX^4/4}$. Its fourth moment tends to the Gaussian value 3 as $g \downarrow 0$ by dominated convergence. Equation (17) gives the stated limit. □

This is a finite nonlinear example in which integrating fluctuations generates a restoring quadratic term at a classically flat point. The local frequency associated with the defined coarse potential is $\omega_{\text{cell}}^2 = \chi_\hbar/(mw)$, so

$$\omega_{\text{cell}}^2 \sim \frac{3\lambda\hbar}{ma} \quad (g \downarrow 0). \tag{18}$$

The word frequency here describes this coarse-potential diagnostic; it is not an asserted spectral gap or a real-time oscillation frequency without a reconstruction argument.

A direct insertion calculation verifies the generated term. At fixed $a, w, \hbar, \mu$, the right derivative at $\lambda = 0$ is

$$\partial_\lambda A_\hbar(\mu)|_{0+} = \frac{w}{4}\left[\mu^4 + 6\mu^2\frac{\hbar}{a} + 3\left(\frac{\hbar}{a}\right)^2\right]. \tag{19}$$

This follows by differentiating (3) under the Gaussian integral. It is a statement about a derivative, not a claim that the full perturbation series converges. It agrees with the coefficient 3 in (18).

5. One nonlinear halving and its large-field events

In a finite nearest-neighbour Euclidean time chain with fixed endpoint positions, use the normalized joint law and choose a set $\mathcal{I}$ of interior vertices with no two adjacent, retaining all other vertices. Conditioned on the retained values, the variables Z_j, $j \in \mathcal{I}$, are independent, each with a density of the form (3), with its own a_j, w_j, μ_j. The exact retained kernel remains a product of nearest-neighbour kernels, although their actions need not have the original kinetic-plus-potential form.

Define a bad event relative to each cell's conditional minimizing value:

$$B_j = \left\{|Z_j - z_{*,j}| > b\sqrt{\hbar/a_j}\right\}, \qquad p_b = 2e^{-b^2/2}. \tag{20}$$

For any $H \subset \mathcal{I}$, Proposition 1 and conditional independence give, including after averaging the retained variables,

$$\mathbb{P}\left(\bigcap_{j\in H} B_j\right) \le p_b^{|H|}. \tag{21}$$

For $b > \sqrt{2\log 2}$ this is a decaying bound. It controls the normalized nonlinear measure, including configurations outside the chosen tubes. It is one halving statement for this convex local action. After further halvings the generated interactions must be controlled; we have not proved that the same original-potential formula closes.

6. Written controls and physical meaning

For $V = 0$, (3) gives $A_\hbar = 0$ and variance $\hbar/a$, saturating (7). For $V = kz^2/2$, $k \ge 0$, completing the square gives

$$A_\hbar(\mu) = \frac{awk}{2(a+wk)}\mu^2 + \frac{\hbar}{2}\log\left(1 + \frac{wk}{a}\right). \tag{22}$$

Its curvature agrees exactly with (10) and (12); the fluctuation term is independent of μ and leaves that curvature unchanged. This quadratic control distinguishes a force-curvature effect from the quartic fluctuation-generated curvature in section 4.

For $V = -Fz$, (3) gives

$$A_\hbar = -wF\mu - \frac{w^2F^2}{2a}, \qquad \frac{w^2F^2}{2a} = \frac{F^2uvT}{8m}. \tag{23}$$

This recovers the arbitrary-cut Galileo correction in the time-blocking note, with its normalization explicit.

The cell curvature in (16) is a possible scale diagnostic, conditional on the entire weighting model. If a, w, λ are independently fixed and $0 < \chi < a$ is the measured value of the defined cell curvature, inverting (16) gives the model-conditional action interval

$$\frac{a\chi}{3w\lambda} \le \hbar \le \frac{a\chi}{w\lambda(1-\chi/a)^2}. \tag{24}$$

Indeed, with $q = \chi/a$, (16) gives $q/3 \le g$ and $\sqrt{1+4g} \le (1+q)/(1-q)$, so $g \le q/(1-q)^2$. The lower endpoint is asymptotically sharp in the small-g limit. This inference presupposes that the measured response is the response of (3), rather than an unmodeled apparatus effect.

A nonzero measured value could also come from a classical statistical weight with an independently supplied action-like noise parameter. ODE regularity or this positive Euclidean integral alone cannot identify that parameter with the real-time phase constant, a universal apparatus restriction, or Planck's measured value. Those are separate physical tests. The bounded full-cell result is useful because it makes the alternatives quantitatively testable.

7. Consequence for STATE

The first nonlinear time-elimination step is now controlled without replacing its law by a Gaussian: normalized tails, endpoint response, coarse curvature and a simultaneous one-halving bad-event bound are explicit. The convex quartic generates positive flat-centre curvature at supplied $\hbar > 0$, whereas classical elimination remains regular with zero flat-centre curvature. Next investigate branch competition or real-time/record reconstruction of this diagnostic. The Newton necessity question remains active; no positivity of $\hbar$ is derived from the choice of a positive weight. These are written, unrefereed calculations.